

PERFORMANCE TESTING

Performance Testing

Performance testing is carried out to evaluate the responsiveness, stability, reliability, and scalability of the Automated Network Request Management System under different workloads. The testing focuses on the Service Catalog, request submission, record creation, approval workflow, notifications, and Flow Designer execution.

1. Performance Testing Objectives

- Verify system behavior under normal and peak workloads.
- Measure response time for major system operations.
- Verify that requests are processed successfully under concurrent usage.
- Identify performance bottlenecks.
- Monitor system resource utilization.
- Verify workflow and approval execution performance.
- Evaluate the scalability of the solution.
- Ensure that performance does not negatively affect user experience.

2. Performance Test Environment

Component	Configuration
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Platform	ServiceNow
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Application	Automated Network Request Management
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Instance	Personal Developer Instance / Development Instance
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Main Interface	Service Catalog
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Workflow	Flow Designer
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Database	ServiceNow Tables
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Test Users	Multiple simulated/concurrent users
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Test Data	Network request records, users, approvals and catalog variables
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Network	Stable internet connection
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Testing Type	Functional performance, load, stress and concurrency testing
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3. Performance Test Scenarios

ID	Test Scenario	Expected Result
PT01	Open Network Request catalog item	Form loads successfully without noticeable delay
PT02	Submit a Network Request	Request is submitted successfully
PT03	Submit request with all variables	All required information is captured correctly
PT04	Create Network Database record	Record is created successfully
PT05	Execute approval workflow	Approval request is generated successfully
PT06	Approve a request	Workflow proceeds to the next stage
PT07	Reject a request	Rejection path executes correctly
PT08	Send notification	Notification is generated successfully
PT09	Update request record	Record status is updated correctly
PT10	Execute complete Flow Designer workflow	End-to-end workflow completes successfully
PT11	Multiple users submit requests simultaneously	System processes concurrent requests correctly
PT12	Multiple approval requests	Approval workflow remains stable
PT13	Search and filter request records	Results are returned correctly
PT14	Open request records repeatedly	Records load consistently
PT15	High-volume request submission	System remains stable under increased workload

4. Performance Metrics

Metric	Description
Response Time	Time taken by the system to respond to a user action
Throughput	Number of requests processed during a specific period

Metric	Description
Concurrent Users	Number of users performing operations simultaneously
Error Rate	Percentage of failed transactions
Workflow Execution Time	Time required to complete the automated workflow
Database Processing Time	Time required for record operations
Notification Processing Time	Time required to generate/send notifications
CPU Utilization	Processor utilization during testing
Memory Utilization	Memory consumption during testing
SLA Performance	Ability of the system to process requests within defined SLA targets

5. Load Testing

Load testing evaluates the application under expected user workloads.

Load Level	Example Users	Purpose
Light Load	10	Verify basic system performance
Normal Load	25	Represent normal usage
Heavy Load	50	Evaluate increased workload
Peak Load	100	Evaluate peak concurrent usage
Stress Load	150+	Identify system limits and degradation

These values are **test-planning examples** and should be changed according to the actual expected users of the deployed system.

6. Concurrency Testing

Concurrency testing verifies whether multiple users can perform operations at the same time without data loss or workflow failure.

Examples:

- Multiple users submitting Network Requests simultaneously.
- Multiple requests entering the approval process.

- Multiple records being created simultaneously.
- Multiple notifications being triggered.
- Multiple IT team members updating request records.

Expected Result

The system should process concurrent transactions correctly without duplicate records, lost information, workflow failures, or incorrect status updates.

7. Stress Testing

Stress testing gradually increases the workload beyond normal operating conditions to identify the point at which system performance begins to degrade.

The test should monitor:

- Response time
- Error rate
- Workflow execution
- Record creation
- Database operations
- Notification processing
- Resource utilization

The purpose is to identify system limitations and determine whether additional optimization or configuration is required.

8. End-to-End Performance Test

The complete request lifecycle can be tested as one transaction:

Submit Request → Capture Variables → Create Record → Approval → Flow Logic → Fulfillment → Notification → Update Record → Closure

The project workflow already follows these major automation stages through Flow Designer.

9. Performance Test Case Template

Test Case ID	Operation	Input	Expected Result	Actual Result	Status
PT01	Open Catalog	Network Request	Form opens successfully	To be measured	Pending
PT02	Submit Request	Valid request details	Request submitted	To be measured	Pending
PT03	Create Record	Submitted request	Network record created	To be measured	Pending
PT04	Approval	Valid request	Approval generated	To be measured	Pending
PT05	Approve	Approval request	Workflow continues	To be measured	Pending
PT06	Reject	Approval request	Rejection path executes	To be measured	Pending
PT07	Notification	Approved/Rejected request	Notification generated	To be measured	Pending
PT08	Update Record	Processed request	Status updated	To be measured	Pending
PT09	Concurrent Requests	Multiple users	Requests processed correctly	To be measured	Pending
PT10	End-to-End Flow	Complete request	Workflow completes successfully	To be measured	Pending

10. Performance Acceptance Criteria

Parameter	Suggested Target
Request Submission	Successful without transaction failure
Record Creation	Successful for valid requests
Approval Processing	Correct approval/rejection path

Parameter	Suggested Target
Notification	Generated at configured stages
Error Rate	As low as practically achievable
Data Integrity	No missing or duplicated request data
Concurrent Processing	No workflow or record corruption
Workflow Completion	Successful end-to-end execution
Scalability	Stable performance as workload increases

The exact numerical response-time and resource-utilization targets should be defined based on the actual ServiceNow environment and organizational SLA requirements rather than assumed values.

11. Performance Testing Process

Identify Requirements → Define Test Scenarios → Prepare Environment → Prepare Test Data → Execute Tests → Monitor Metrics → Analyze Results → Identify Bottlenecks → Optimize → Retest → Document Results

12. Expected Outcome

Performance testing should demonstrate that the Automated Network Request Management System can reliably handle network-request operations while maintaining acceptable responsiveness and data integrity. It should also identify potential bottlenecks and provide a basis for future optimization and scaling.

The project's intended automation of request processing, approvals, notifications, and record updates supports faster and more efficient request management.

Planning Logic

The planning logic defines the step-by-step approach to design and implement the Automated Network Request Management System in ServiceNow. Each step ensures smooth flow from request submission to fulfillment and closure.

Step	Planning Logic Description
1. Identify Need	Identify the problems in the existing network request process.
2. Define Requirements	Determine the information, approvals, records, and notifications required.
3. Design Request Form	Create the Network Request catalog item and configure the required variables.
4. Create Data Structure	Create the Network Database table and required fields to store request information.
5. Automate Workflow	Build the Flow Designer workflow to process submitted requests automatically.
6. Apply Approval Logic	Route requests for approval and handle approved or rejected outcomes.
7. Fulfill Request	Create the required record/task and allow the IT/Network team to process it.
8. Notify Users	Send notifications regarding approval, processing, and request status.
9. Update & Close	Update the request record and close it after successful fulfillment.
10. Test & Validate	Test the complete workflow using sample requests and verify each stage.

